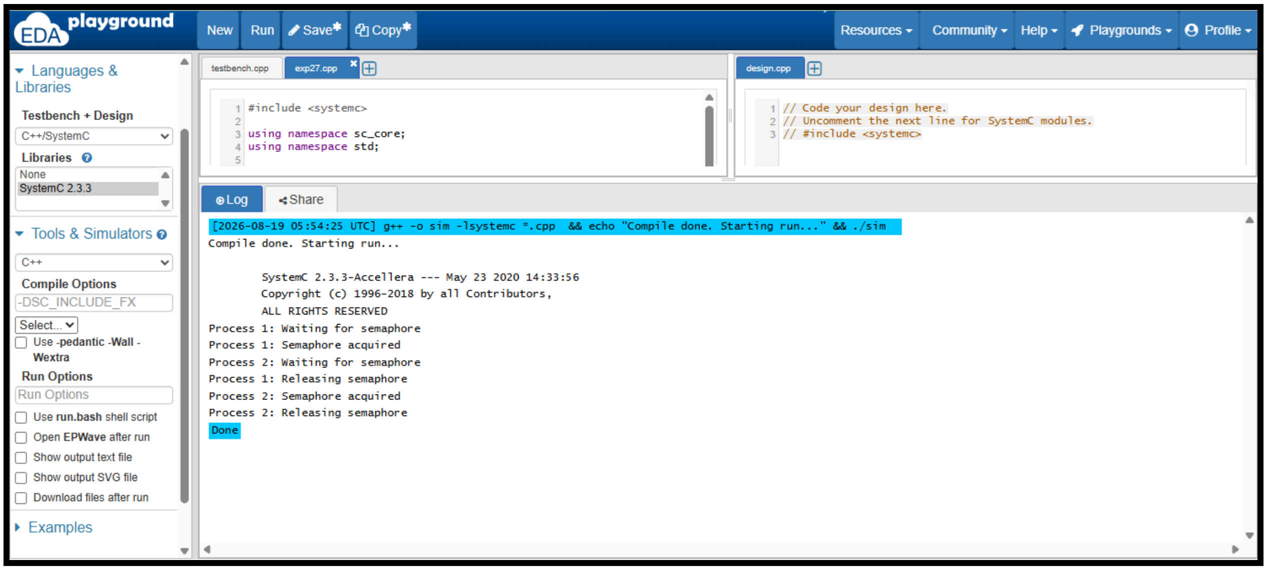

C++ Program:

```
#include <systemc>

using namespace sc_core;
using namespace std;

SC_MODULE(BinarySemaphore)
{
    sc_semaphore semaphore;

    SC_CTOR(BinarySemaphore)
        : semaphore("semaphore", 1)
    {
        SC_THREAD(process1);
        SC_THREAD(process2);
    }

    void process1()
    {
        wait(5, SC_NS);

        cout << "Process 1: Waiting for semaphore"
            << endl;

        semaphore.wait();

        cout << "Process 1: Semaphore acquired"
            << endl;
```

```
        wait(10, SC_NS);

        cout << "Process 1: Releasing semaphore"
            << endl;

        semaphore.post();
    }

    void process2()
    {
        wait(6, SC_NS);

        cout << "Process 2: Waiting for semaphore"
            << endl;

        semaphore.wait();

        cout << "Process 2: Semaphore acquired"
            << endl;

        wait(10, SC_NS);

        cout << "Process 2: Releasing semaphore"
            << endl;

        semaphore.post();
    }
};


int sc_main(int argc, char* argv[])
{
    BinarySemaphore b("BinarySemaphore");

    sc_start(30, SC_NS);

    return 0;
}
```